Class-Asymmetric Robustness and Transformation-Order Effects in Lightweight Audio Deepfake Detection

Abstract

We examine how channel augmentation changes audio deepfake detection under ordered noise and Opus processing. Three paired MobileNetV3Small–LSTM seed blocks share 640 ASVspoof 2019 LA training recordings, within-seed initialization, source order and clean-development checkpoint selection. On shared development sources, macro stressed ROC-AUC increased in all three pairs, with a mean paired gain of 0.1880 and sample standard deviation of 0.0562. Stressed FAKE misses fell in all three pairs at threshold 0.5. REAL false alarms varied by seed and stratum, and A05/A06 stress misses increased in the first two pairs and decreased in the third. A separately locked within-release evaluation covered 1,088 recordings and only the original four models. Its two fitted pairs improved stressed ranking and reduced pooled FAKE misses, while clean FAKE misses and stressed misses on A17–A19 increased. Reversing noise and Opus on the same source changed classwise decision transitions. The results support reporting ranking, false alarms and misses jointly and evaluating relevant processing orders on matched recordings.

Keywords. Audio forensics, channel augmentation, codec robustness, spoofing countermeasures

1 Introduction

Audio deepfake detectors are used to flag speech that may have been synthesized or voice converted. In an authentication workflow, a false alarm can divert a bona fide recording to manual review, whereas a missed spoof can leave impersonation undetected. The cost of either decision depends on the workflow and the prevalence of spoofed traffic. A detector chosen by aggregate discrimination alone can behave differently at a fixed operating point after a recording passes through ordinary signal processing.

Noise and lossy coding are particularly relevant because they can occur in either order. Speech recorded in a noisy room may be encoded before transmission, while background audio can also be introduced after decoding. These pipelines preserve the intended class of a source, but change the evidence presented to a detector. Even with identical noise and codec settings, reversing the operators can change the waveform, spectral features, ranking, and fixed-threshold decisions. Measurements across matched views can isolate the influence of the implemented signal-processing order within a recorded source.

The accepted LAVA conference manuscript benchmarks six heterogeneous detectors. The extension presented here addresses a different question through a matched intervention on one detector architecture. Within each of three initialization blocks, the training recordings, order, optimizer, numerical backend, and checkpoint rule are shared. The controlled factor is whether each training recording is presented clean or with a view sampled from a channel bank. Both fitted models within each block are evaluated on each development recording and condition. This enables direct accounting of ranking changes, class-specific error changes, and which sources cross the same decision threshold.

The analysis asks whether aggregate robustness moves in the same direction across the three blocks, which attacks and speakers bear the associated errors, and how those errors change under reversal of a conventional codec and noise chain. The resulting evidence is a controlled

comparison and a source-linked diagnostic. Sections 2–5 specify the design, and Sections 6–8 distinguish measured findings from possible explanations and threats to validity.

1.1 Operational question and scope

The action following a detector decision determines which error is costly. A system that routes suspicious clips to an analyst incurs more review when authentic audio crosses the threshold. A system used to prioritize potentially manipulated evidence has a different cost for missing a synthetic or converted clip. A balanced validation cohort therefore answers a diagnostic question about conditional errors. It does not estimate workload for a deployment with an unknown spoof prevalence. The operational quantities reported here are denominated within class and condition, so that a practitioner can consider the two errors separately.

Clean evaluation provides a useful reference point because a detector can rank bona fide and spoofed recordings well before channel processing changes their score distributions. In audio production and communications, however, noise and compression can occur in either order. It is possible to keep the source, nominal noise level and codec settings fixed while varying the order. An observed score change then identifies sensitivity to the specified composition on that recording. Comparison across two different recording lists would entangle this effect with differences in speakers, content and synthesis examples.

The training comparison addresses a complementary question. Exposure to individual channel views may help on some transformed evaluation views even when the specific compositions were absent from the training-view bank. The experiment measures this transfer within ASVspoof 2019 LA under matched training and checkpoint selection. It records whether the rank benefit and fixed-threshold behavior agree, and whether the observed differences survive inspection at the attack and speaker level. This scope places claims about ordered signal processing and model response on separate evidential footing from claims about deployment or forensic cue preservation.

1.2 Research gap and questions

RawBoost and ADD-C establish practical approaches to channel robustness, and bona fide cross-testing shows that authentic speech support matters for reported errors [1, 2, 3]. Three questions remain open for the matched intervention and channels examined here.

1. **Ranking and operating risk (RQ1).** How does replacing clean training views with sampled channel views affect discrimination, false alarms and misses under clean and distorted evaluation? ROC-AUC compares ranks over possible operating points. Classwise errors describe decisions at a given threshold. The distinction is standard, and the open empirical question is when the two assessments agree under matched training. We address it with paired ROC-AUC and fixed-threshold contrasts, followed by a descriptive threshold sweep.

2. **Unequal robustness (RQ2).** Which classes and attack IDs improve or deteriorate under the same intervention, and how consistent are the observed changes across seeds and conditions? Prior domain studies motivate separate class and subgroup analysis [3, 4].

Table 1: Scientific extension of the accepted conference study. Each column describes its own experimental population.

Dimension	Conference study	Journal extension
Question and models	Offline benchmark of six available detector artifacts with different training histories	Controlled ERM and channel-view comparison on MobileNetV3Small–LSTM, three paired development seeds
Cohort and stress	2,737 checksum-disjoint clean test recordings and 100 stress recordings under nine conditions	ASVspoof 2019 LA train and development, plus 1,088 locked held-out sources and paired noise/Opus order
Inference and uncertainty	Paired clean correctness, Holm-adjusted McNemar tests and stratified bootstrap	Class-conditional risk, attack recovery and harm, speaker-cluster intervals within each seed
Compute and scope	Historical desktop inference timing	Training logs and bounded CPU inference audit, with one architecture and within-release validation

The present source linkage distinguishes shared errors, recoveries and new errors. Speaker intervals characterize each fitted comparison. Three seed blocks and limited subgroup support constrain generalization.

3. **Processing order (RQ3).** How does reversing additive noise and Opus coding on the same recording affect REAL and FAKE score distributions and decisions? Noncommutativity is an established property of signal-processing compositions. Same-source pairing holds speech content, class and nominal transform settings fixed and shares the noise realization, so changes can be attributed to the tested pipeline order within each recording. Achieved distortion can differ, and this design does not identify an independent forensic cue.

1.3 Main contributions

1. **Controlled evaluation of channel augmentation.** Three paired MobileNetV3Small–LSTM blocks share architecture, within-seed initialization, source order, optimization and the registered clean-development checkpoint rule. We report ranking and classwise decisions on development. The original two pairs also have a locked within-release held-out evaluation.

2. **Class-asymmetric and attack-specific robustness.** We characterize false alarms and misses, including A05/A06 on development and A17–A19 on held-out, with source-level recovery and deterioration and speaker-aware uncertainty. The threshold sweep and additional subgroup decompositions are retrospective.

3. **Transformation-order sensitivity.** Paired Noise10–Opus24 and Opus24–Noise10 views identify score changes and classwise decision transitions on the same sources. Matched standalone components help delimit the observed response to these operators.

2 Related Work

2.1 Detector architectures and benchmark design

RawNet2 extracts representations directly from speech waveforms, while AASIST uses graph attention over spectro-temporal representations [5, 6]. A compact convolutional backbone offers a different compute and representation trade-off. MobileNetV3 was designed for mobile visual recognition [7]. Here, it encodes successive Mel-derived feature segments and a recurrent head aggregates them in time. Published lightweight alternatives such as DIN-CTS combine a different feature extractor with contrastive training [8]. The historical detectors provide context for this choice, and the controlled intervention uses the same backbone and head in every arm.

ASVspoof 2019 distinguishes logical-access synthetic or converted speech from physical-access replay [9]. The 2021 challenge extends benchmarking to separate logical-access, physical-access, and deepfake tasks under different conditions [10]. ASVspoof 5 scales crowdsourced collection and expands spoofing and adversarial evaluation [11]. In-the-Wild evidence shows why a benchmark split cannot by itself establish transport to collected real-world material [12]. The present development subset has speaker separation from training and includes attack IDs seen in training. In the within-release evaluation, A16 and A19 share algorithms with A04 and A06, respectively, and A17 shares a framework with A05 [9]. Those relationships constrain any interpretation of held-out attack IDs.

The multilingual MLAAD resource targets variation in synthesis systems and languages [13]. Its design provides a complementary test of transport to language and generation domains beyond this within-release comparison.

2.2 Signal transformations and robust learning

Opus is a conventional lossy codec [14]. Compression can change spectral and temporal cues through content-dependent coding decisions. Resynthesis with a learned neural codec is a different transformation family, with different artifacts. A review and case study by Maltby et al. place codec effects alongside dataset bias and other failure modes [4]. These studies motivate treating the nominal operator, signal representation, and downstream detector as separate factors.

RawBoost combines convolutive, impulsive, and stationary waveform perturbations and reports countermeasure gains on ASVspoof 2021 LA [1]. The present augmentation instead samples one registered gain, noise, codec, or clean view for each visit and uses ordinary binary cross-entropy. Frequency-selective adversarial training changes how perturbations are chosen [15]. Group distributionally robust optimization places weight on high-risk groups [16], and DK-CAST adds supervision through clean-to-degraded distillation [17]. SONAR evaluates spectral-contrastive residual representations under diverse shifts [18]. These approaches address changes to objectives or representations. The present experiment holds the representation family and objective fixed to estimate the effect of the view policy.

2.3 Order effects and operating-point uncertainty

ADD-C tests detection under conventional codecs and simulated packet loss and evaluates augmentation for those communication conditions [2]. Bona fide cross-testing varies authentic-

Table 2: Comparison of research designs. Different cohorts and operating rules prevent a direct performance ranking.

Study	Data or evaluation setting	Treatment or model	Relationship to the present comparison
RawBoost and F-SAT [1, 15]	ASVspoof and DeepFakeVox-HQ, respectively	Waveform augmentation and frequency-selective adversarial training	Establish robust training precedents. Neither is retrained as a matched arm here.
DK-CAST and SONAR [17, 18]	Speech spoofing and domain-shift evaluations	Distillation and spectral residuals	Intervene on learning objectives or representations. The present study intervenes on training views.
ADD-C and bona fide cross-testing [2, 3]	Simulated communication and diverse authentic speech	Codec and loss grids, and bona fide domain testing	Motivate stress coverage and authentic-speech reporting. The present study controls order on a fixed source.
Present study	Three development initialization blocks, shared development and locked within-release held-out	Matched lightweight detector, ordinary augmentation	Links ordered processing and class-specific error transitions on the same source. Broader transport and mechanisms remain open.

speech domains against spoofed sets and shows why authentic-speech coverage matters [3]. Both motivate class- and condition-specific evaluation. Here the comparison reverses noise and conventional Opus on the same source with a shared noise plan, and measures how a matched training-view intervention changes the resulting threshold errors. This narrower intervention yields source-level evidence about the two orders under the tested settings.

Calibration concerns the interpretation of scores, while selective prediction introduces a reject action [19, 20]. Work on calibrated audio deepfake detection illustrates the importance of these distinctions in the speech domain [21]. Here the sigmoid output is denoted by p for convenience and used as an uncalibrated score. A rank improvement and an error increase at a fixed threshold are compatible. Table 2 summarizes how the nearest work differs in treatment and evaluation scope.

2.4 Channel exposure and representation choices

Work with a pretrained wav2vec 2.0 front end and data augmentation provides published evidence for a different route to addressing channel and source shift [22]. The present comparison uses a compact Mel-sequence classifier with the same representation family in both arms. This limits the treatment comparison to exposure during fitting and avoids interpreting a change of backbone, input bandwidth, or pretraining corpus as an augmentation effect. The design also avoids comparisons with a published EER obtained under a different source partition and operating protocol.

Cross-domain multilingual benchmarks such as XMAD-Bench address variation across syn-

thesis and language domains [23]. Such benchmarks answer a different question from a source-matched transformation test. Pairing holds a source fixed across channel operations, improving interpretability of the channel contrast. It also conditions the result on the source population in the selected cohort. A cross-domain benchmark can broaden the population but introduces additional differences that must be accounted for when attributing a score shift. Both forms of evidence are relevant to deployment, and neither can replace the other.

Group-robust learning emphasizes a specified subgroup criterion and usually requires a choice of groups and weights. A standard augmentation objective instead averages the loss across sampled exposures. This distinction matters for an attack with a small number of training sources or a poorly represented channel interaction. The latter objective need not protect the worst subgroup even if it improves aggregate discrimination. The journal compares the ordinary view-policy intervention and records subgroup effects. No class-aware weighting, consistency loss, or group-robust optimizer is introduced.

The prior designs also differ in the statistical question posed by the operating rule. EER estimates an equal-error summary from the evaluation score distribution, whereas a fixed boundary permits separate estimates of false alarms and misses under each processing condition. We therefore keep the registered boundary and report a retrospective curve analysis around it. The attack and speaker contrasts are conditional on fitted models. They complement benchmark-level results without equating a new source population, a different intervention, or a different operating rule.

3 Problem Formulation

Let x_i be source recording i, g_i its speaker, and $y_i \in \{0, 1\}$ its label, with 0 denoting REAL and 1 denoting FAKE. Let $p_\theta(Tx_i) \in [0, 1]$ be the detector score after transformation T. The decision $\widehat{y}_{i,T} = \mathbf{1}\{p_\theta(Tx_i) \geq 0.5\}$ fixes the operating point. The class-conditional risks in a fixed cohort and condition are

$$R_0(T) = \frac{\sum_{i:y_i=0} \widehat{y}_{i,T}}{n_0}, \qquad R_1(T) = \frac{\sum_{i:y_i=1}(1 - \widehat{y}_{i,T})}{n_1}. \tag{1}$$

They correspond to false alarms and spoof misses. For a set of stress conditions, an error count adds the numerator across conditions and uses n_y times the number of conditions as the denominator. It describes risks across repeated views.

Receiver operating characteristic area under the curve (ROC-AUC) compares the ordering of FAKE and REAL scores, crediting half for ties. A monotonic score transformation can preserve ordering and change the number of scores crossing 0.5. Equal error rate (EER) is summarized at the empirical ROC point closest to equality between false-alarm and miss rates, using their mean at that point. Both metrics require support from both classes. The stress macro ROC-AUC is the equally weighted mean of ROC-AUC computed separately for the three registered stress conditions.

Let $N_i(x) = x + \nu_i$ denote insertion of the registered fixed-absolute noise for source i, and let C denote encoding and decoding with Opus. Define $T_{NC} = C \circ N_i$ and $T_{CN} = N_i \circ C$. Because C uses content-dependent analysis and lossy coding, in general $C(x + \nu_i) \neq C(x) + \nu_i$. In one

case the codec processes the added noise, while in the other it does not. The contrast

$$d_i = p_\theta(T_{NC}x_i) - p_\theta(T_{CN}x_i) \tag{2}$$

measures a directional score response. A decision transition records which order alone causes an error, if either does. The contrast can reflect altered input features, score placement or cues used by the detector. An independent cue measure would be needed to interpret it as forensic-evidence retention.

3.1 Ranking and conditional risk estimate different quantities

For scores S_1 and S_0 drawn from the FAKE and REAL groups of a condition, the population form of the ranking estimand is

$$\mathrm{AUC}(T) = \Pr(S_1 > S_0) + \tfrac{1}{2}\Pr(S_1 = S_0). \tag{3}$$

The finite-sample estimate averages this comparison across class pairs [24]. Each pair contributes to ranking even when both scores lie on the same side of 0.5. Conversely, a source near 0.5 may change its decision under a small perturbation that changes almost none of the pairwise comparisons. Thus AUC and threshold errors can move in different directions. The macro stress AUC gives each of the three stress conditions equal weight. Pooling scores before ranking would also compare scores across different conditions and would estimate a different quantity.

Let $D_i^{(m)}(T)$ be the error indicator under method m. The paired contribution to a conditional error difference is $D_i^{(\mathrm{channel})}(T) - D_i^{(\mathrm{ERM})}(T)$. Its values are $+1$ for a newly introduced error, -1 for a recovered decision, and zero for agreement in error status. Summing these contributions gives the net count while retaining the ability to distinguish which sources changed in each direction. This matters when equal error counts conceal disjoint sets of failures. For an attack group, the estimand is the miss rate conditional on that FAKE group. Ranking within a single FAKE group against itself has no binary detection interpretation.

3.2 Order effects and component diagnostics

For a frozen model, let $p_0, p_N, p_C, p_{NC}, p_{CN}$ denote scores from the clean, noise-only, codec-only and two composed inputs. The signed order contrast is $p_{NC} - p_{CN}$. A complementary descriptive contrast is

$$h_i = \tfrac{1}{2}(p_{NC} + p_{CN}) - p_N - p_C + p_0. \tag{4}$$

This contrast is zero if the two component changes add on the score scale and the two orders have the same score. Departures may also arise from nonlinear features or the sigmoid score mapping. Consequently the contrast is a diagnostic of output nonadditivity, and no inference about which speech cue was preserved follows from its magnitude. The standalone matched components are available for 9107 and 9207. The registered score and decision order contrasts are available for all three development blocks.

3.3 Decision costs and attack mixtures

Suppose a deployment has spoof prevalence π and costs c_0 and c_1 for a REAL false alarm and FAKE miss, respectively. For a fixed channel and threshold, a simple expected error cost is

$$(1 - \pi)c_0 R_0(T) + \pi c_1 R_1(T). \tag{5}$$

The prevalence and costs are not estimated here. The expression clarifies why neither class-balanced training nor an equal-weight stress macro average specifies an operational decision rule. The 0.5 rule provides a fixed point for a matched comparison and leaves calibration and cost-optimal threshold selection for a separate study.

The attack mixture also affects the observed miss rate. For represented attack IDs a with class-conditional weights w_a, $R_1(T) = \sum_a w_a R_{1,a}(T)$. Changing these weights changes the aggregate even if all attack-specific rates stay fixed. In development the mixture comes from the selected sources, and repeated views preserve their attack identity. The reported aggregate therefore pertains to this mixture. Separate attack rows enable comparison without transferring these mixture weights to another population.

3.4 Matching the intervention and defining the comparison

Let b index a training seed block, m index the two fitting policies, and $s_{ibm}(T)$ denote the FAKE score from the selected checkpoint for recording i under condition T. A between-arm score difference changes m while preserving i, b, and T. A within-model order difference instead changes T while preserving i, b, and m. The first comparison concerns the total effect of the fitting policy followed by checkpoint selection. The second concerns the detector response to a specified input manipulation. Interpreting one as the other would conflate changes in learned weights with changes in input.

A threshold transition can be analyzed without introducing a second operating point. For a REAL source, crossing from below to above 0.5 introduces a false alarm. For a FAKE source, the same crossing repairs a miss. A directional shift can therefore move the two risks in opposite directions even when its sign is common across classes. The empirical question is where each distribution lies relative to the fixed threshold and how its ordering changes relative to the other class.

4 Matched Training Methodology

4.1 Detector and objective

Audio is decoded as mono at 22.05 kHz with a three-second limit and processed using the frozen P0 input contract. Six consecutive 0.5 s windows are mapped to log-Mel power with 128 filters, a 2,048-sample Hann window and a 512-sample hop. P0 floors the power at 10^{-10}, references each segment to its maximum, clips the dynamic range to 80 dB, and uses bilinear resizing and three-channel replication to produce 224-by-224 tensors. An ImageNet-initialized MobileNetV3Small backbone encodes each segment. An LSTM with 128 units aggregates the

feature sequence before a binary sigmoid head. The sealed experiment identifies the architecture and input implementation.

The backbone omits the image-classification top and uses spatial average pooling. It applies shared weights across the segment sequence. The temporal layer returns the final representation, which feeds a 64-unit rectified-linear layer and a dropout layer with rate 0.4, followed by a single sigmoid output. During partial fine-tuning, the final 20 backbone layers are eligible for updates, with batch-normalization layers frozen. During warm-up, the backbone is frozen throughout. These settings define the fitted detector and are independent of the checkpoint I/O and resume mechanism.

Both training arms minimize mean binary cross-entropy on one view per source at each visit. Read-only inspection of the sealed exports found 213 stored tensors with 1,308,401 scalar values per model. The count includes nontrainable state and does not estimate the number of variables updated in either fine-tuning stage. If V_i is the view and $q_i = p_\theta(V_i x_i)$, then

$$\mathcal{L}(\theta) = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log q_i + (1 - y_i)\log(1 - q_i)]. \tag{6}$$

The empirical risk minimization (ERM) arm sets V_i to the clean view. The channel arm samples from clean, half gain, 10 dB noise, and 24 kb/s Opus. Neither ordered stress chain is a supervised training view. Training-view randomness is determined by the registered seeded procedure, while the source order and initialization are identical within each pair.

Each completed arm logged 13,440 source visits. In the order clean, half gain, Noise10, and Opus24, the realized channel counts were 3,293, 3,297, 3,349, and 3,501 for seed 9107, 3,403, 3,415, 3,338, and 3,284 for seed 9207, and 3,325, 3,384, 3,320, and 3,411 for seed 9307. ERM used the clean view at every visit. These counts describe the completed budget. The clean-selected epochs differ between arms, so the selected models use different prefixes of those visits. The comparison estimates the view policy as a bundle, including reduced clean exposure and its interaction with checkpoint selection.

4.2 Optimization and model selection

Adam [25] is used for 840 updates over 21 epochs. Each update accumulates gradients over 16 single-source microbatches. One warm-up epoch trains the head, followed by 20 partial fine-tuning epochs with frozen batch-normalization state. The respective learning-rate stages start at 0.001 and 0.0001 and use the registered linear decay. The registered CPU engine uses full-shape recomputation and a fixed numerical contract. Checkpointing and deterministic resume support reproducibility.

Validation tier A evaluates clean mixed-class development at each epoch. Tier B also measures the registered stress and REAL-only endpoints every third epoch and at completion. The minimum clean A BCE checkpoint is chosen, and tier C evaluates its complete development grid. Risk guards are evaluated as a comparison of the selected checkpoints. They do not select a different epoch or alter the fixed decision rule.

Figure 1 shows the observed clean validation trajectories. All runs completed epoch 21, although the selected checkpoints fall between epochs 13 and 19. The curves compare the same

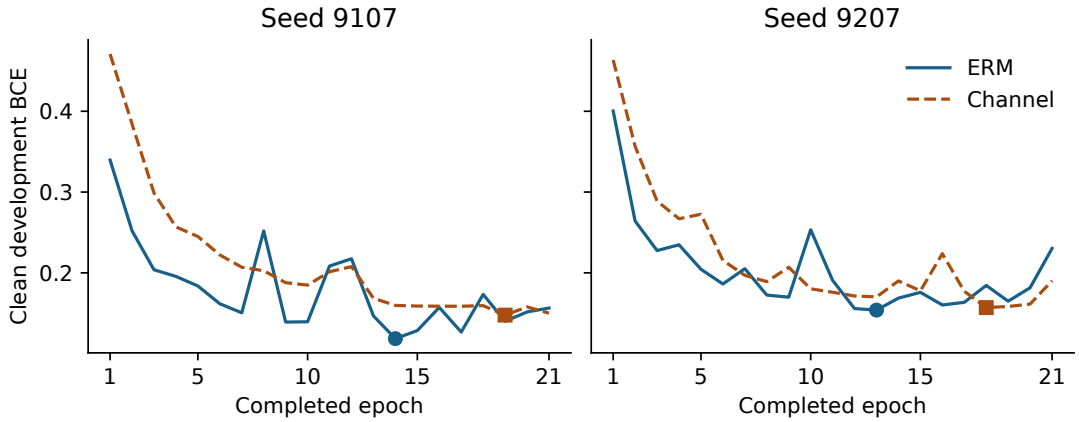


Figure 1: Recorded clean development BCE over 21 epochs for both methods and seeds. Each observation uses the same 224 development A recordings. Markers show the selected epoch for each model.

validation loss on the same cohort. Training losses reflect different view policies and are excluded from the plot.

4.3 Treatment, numerical contract and reproducibility

The two treatment arms use the same binary labels and classification objective. A change of view preserves the source identifier and training label. This keeps the number of optimizer updates and the number of source visits identical within a seed while exposing the channel arm to nuisance variation. The clean, gain, noise and codec views share the preprocessing policy. Each arm learns its own weights from its assigned views. The comparison estimates the total effect of the registered augmentation treatment and subsequent selection rule, so it includes changes in which epoch is selected.

The frozen batch-normalization policy avoids recomputing running statistics from different mixtures of training views. This controls one source of between-arm variation but also limits adaptation to the new input distribution. The warm-up and partial fine-tuning stages use the same schedule and update accounting in both arms. The optimizer accumulation unit is the committed update. Checkpoints store numerical state and the source cursor so that interrupted execution can resume without changing the registered sequence.

At inference, the selected checkpoint is restored from the sealed artifact and receives each registered view through the same P0 feature contract. P0 combines a power floor, per-segment peak reference, dynamic-range clipping, interpolation, and channel replication. Although the peak reference reduces uniform gain sensitivity over the linear range, near-floor values, padding, clipping and interpolation can still matter. Identical preprocessing between arms makes the treatment comparison interpretable, and leaves open how the effect would change with a different feature pipeline.

The analysis records a score and class decision for every model–source–condition cell. The source and condition identity must match before paired results are computed. A missing or changed row would invalidate a paired difference. This accounting also separates the final validation of the selected checkpoint from the checkpoints used to monitor optimization. The

validation record reported by a live monitor can describe a different epoch from the exported model.

4.4 Why selection is part of the intervention

Clean-loss selection imposes a common rule, while allowing the selected epoch to depend on the training policy. A comparison at the same update would answer a separate optimization question. In the deployed comparison each arm is evaluated at its own clean-selected checkpoint, with the threshold held fixed and stress scores excluded from checkpoint choice. Reporting the selected updates makes the distinction visible and prevents accidental substitution of final weights.

The identical source order prevents a difference in source scheduling from explaining an observed arm contrast within a block. Channel training consumes stochastic view choices, while the ERM arm applies the identity view. This difference is the intervention. The two transformations used in the ordered stress chains are evaluated in composition even though the composed views are absent from the training-view bank. Consequently, improvement on a composition measures transfer across views within the tested cohort and is conditional on the chosen model family and fitting budget.

5 Experimental Setup

5.1 Cohorts and conditions

The resource-bounded medium cohort includes 640 ASVspoof 2019 LA training recordings, balanced between REAL and FAKE, and attack IDs A01–A06. Development A consists of 112 REAL and 112 FAKE recordings from seven speakers. Development B adds 144 REAL-only recordings from nine speakers. The registered audit checks disjoint training and development speaker IDs. The same development sources are used for the three independently initialized blocks, so the blocks measure sensitivity to training, rather than to a new test population.

The seven scored views per model are clean, two gain controls, Noise5, Opus64, and the two reversed chains. The stress endpoint uses Noise5 and the chains. In the chains, a fixed-absolute 10 dB noise plan and the 24 kb/s Opus encoder are shared across order. Noise5 and Opus64 are separate conditions. The matched standalone Noise10 and Opus24 diagnostic was performed for seeds 9107 and 9207. Seed 9307 supplies composed-order development scores only. Table 3 summarizes the fixed design.

The medium cohort was chosen to permit matched CPU runs with balanced class support. It has 320 REAL and 320 FAKE sources. The full training partition contains 2,580 REAL and 22,800 FAKE recordings, so a comparison with that partition would change both training size and class mixture.

5.2 Paired uncertainty and efficiency

For each seed, differences are channel minus ERM. Percentile intervals use 10,000 paired speaker-cluster bootstrap draws with seed 91109. A and B are sampled separately. Within a sampled speaker, all sources and their views retain their paired model scores. This preserves source and

Table 3: Registered prospective development experiment.

Factor	Specification
Training	640 sources, 320 per class
Development	A 224 mixed-class, B 144 REAL
Blocks	Matched development seeds 9107, 9207 and 9307
Architecture	MobileNetV3Small with LSTM128
Budget	21 epochs, 840 updates per arm
Selection	Minimum A clean BCE
Operating point	FAKE when score is at least 0.5
Outputs	2,576 selected-model scores per arm

speaker dependence and avoids treating transformed views as independent trials. Mean, sample standard deviation and range across the three blocks are descriptive. They do not estimate a population confidence interval over random initializations.

The six runs used a Windows CPU environment. Optimizer and validation durations come from run logs and may include repeated work after interruption. The registered evaluation policy preserves the 0.5 threshold and excludes calibration from model selection.

5.3 Design controls and support

Development A and B play different roles. A supplies both classes for ROC-AUC, EER and fixed-threshold risks. B adds a second, REAL-only speaker stratum for false-alarm sensitivity. Pooling B into A would change the implicit weighting of speakers and the set of REAL–FAKE pairs. The two strata are retained separately. Counts in the mixed-class stratum are supported by 112 original recordings per class. The three stress views contribute 336 outcomes per class but still derive from those same recordings.

Within each seed block, both fitted models see the same registered development sources and transformed waveforms. Between seeds, the independent initializations vary the fitted functions, but the evaluation cohort stays fixed. Thus there are two kinds of pairing. Source pairing controls the input to a between-method comparison, and seed blocking controls initialization and source order during training. All three blocks reuse the same speaker population. All attacks A01–A06 are represented in training. The attack rows therefore show differences across represented attack IDs under stress.

The order experiment inserts the same fixed-absolute noise realization before or after the codec, with the same nominal clean-referenced signal-to-noise ratio. The achieved characteristics after compression need not be equal. Treating achieved SNR or bandwidth as equal would erase part of the intervention. The controls support comparison of those specific pipelines and do not simulate packet loss, device reverberation, proprietary enhancement or a complete communications stack.

5.4 Interpretation of the resampling intervals

Cluster resampling keeps all observations from each sampled speaker together [26]. In each replicate the same speaker weights apply to ERM and channel within a stratum. A speaker

drawn more than once contributes the corresponding repeated cluster weight. The analysis recalculates the contrast under those weights and takes the percentile limits over valid draws. Repeated views of a source and paired arm predictions therefore stay linked. For rank metrics, the class support requirement is evaluated within the resampling design. The evidence records the valid draw count so that interval interpretation can account for any undefined draws.

Seven and nine speakers provide limited empirical support for percentile intervals. Large changes in one speaker can influence the interval and can also affect which attack cases are represented in a resample. The intervals condition on the fitted models and do not incorporate a hypothetical new training seed. The differences across seeds are summarized descriptively, and the secondary attack and speaker comparisons are used to localize effects rather than to claim a family of discoveries.

5.5 Evidence boundaries

The prospective comparison includes the registered score orientation, 0.5 operating rule, fixed budget, clean-loss checkpoint selection and stress conditions. The paired records define the unit for contrasting methods. The separate Noise10 and Opus24 runs for 9107 were added as diagnostics on the same fitted models and development sources. They help characterize departures from additive output behavior, with no independent new-cohort replication. The interpretation of component nonadditivity and nuisance adjustment therefore stays distinct from that of the prospective augmentation comparison.

5.6 Locked Held-Out Evaluation

The primary held-out subset comprises 1,088 recordings from 67 speakers in the official ASVspoof 2019 LA countermeasure evaluation partition [9]. A deterministic hash ranking and speaker round-robin selection with seed 91109 yields 256 REAL recordings and 64 FAKE recordings for each attack A07 through A19. The audit reconciled all 71,237 official trials with the archive and found no shared source IDs, speaker IDs, or audio SHA-256 digests between the selected subset and the registered training, development, and calibration manifests. Complete generator and original-utterance lineage remains unverified.

The lock specifies the four development-selected checkpoints from 9107 and 9207 and four conditions per source: clean, Noise5, Noise10 followed by Opus24, and Opus24 followed by Noise10. All 17,408 predictions were completed. Seed 9307 was evaluated on development only. The sigmoid score is oriented toward FAKE and scores of at least 0.5 are assigned to FAKE. Macro stressed ROC-AUC averages the three stress-condition AUCs. Clean and stressed error rates use class-conditional denominators, and the stressed denominator counts three correlated views per source. Held-out contrasts retain within-seed source pairing and use the same speaker-bootstrap settings on the held-out speakers. Development A and B refer only to the earlier development strata.

All source–condition–model cells were checked against the locked identities. Per-source score shards retain complete model and condition sets, and their hashes were verified before analysis. The registered metrics and uncertainty outputs were recomputed from the committed predictions. The selected sources, models, transformations, and threshold were fixed before scoring. The

Table 4: Development results for all fitted pairs. The selected update identifies the clean-BCE checkpoint. Stressed counts use repeated views.

Seed and arm	Update	Clean AUC	Stress AUC	Stress FA	Stress miss	B clean FA
9107 ERM	560	0.9872	0.6715	54/336	222/336	19/144
9107 CHANNEL	760	0.9814	0.8026	45/336	155/336	31/144
9207 ERM	520	0.9851	0.5946	167/336	143/336	26/144
9207 CHANNEL	720	0.9809	0.7839	70/336	134/336	20/144
9307 ERM	360	0.9822	0.5498	44/336	266/336	21/144
9307 CHANNEL	680	0.9856	0.7934	61/336	119/336	30/144

attack-balanced subset has different support from development A and B, so differences in effect magnitude across partitions are descriptive.

For the ordered views, the source is decoded to 22.05 kHz and truncated at 3 s before applying a common peak limit of 0.25. A source-ID-keyed random draw supplies one fixed-absolute Noise10 vector, scaled against clean reference power. Both chains use this vector. The codec path uses FFmpeg with libopus at nominal 24 kb/s, PCM16 input, timestamp alignment, and decoding back to 22.05 kHz. There is no independent post-channel waveform normalization before P0. The common noise vector does not imply equal achieved post-codec signal-to-noise ratios. The development replay verifies the composed waveforms and score anchors. Held-out evidence uses the locked implementation and recorded scores.

6 Results

6.1 Overall Robustness under Channel Augmentation

Table 4 reports all six development fits. Macro stress ROC-AUC increases by 0.1312, 0.1893 and 0.2436 for 9107, 9207 and 9307, respectively. The mean and sample SD across seeds are 0.6053 ± 0.0616 for ERM, 0.7933 ± 0.0094 for channel, and $+0.1880 \pm 0.0562$ for their paired difference. The clean AUC difference is negative in the first two seeds and positive in 9307. Figure 2 links the two arms within each seed.

Clean A REAL false alarms tie in the first two pairs and increase from 4 to 6 of 112 in 9307. Clean A FAKE misses increase from 6 to 9 in 9107, tie at 9 in 9207, and decrease from 10 to 8 in 9307. Clean B REAL false alarms increase in 9107 and 9307 and decrease in 9207. Table 5 separates within-seed speaker intervals from descriptive variation across fitted pairs.

All three development A macro stress AUC intervals exclude zero. In 9307 the paired interval is [0.1884, 0.3071]. The 9207 stress miss interval includes zero. These intervals quantify speaker resampling conditional on each fitted pair and shared development protocol.

The locked evaluation completed 17,408 predictions for the original four selected models. Table 7 reports aggregate results and Table 8 separates the conditions. Macro stressed ROC-AUC increased in both seed blocks, from 0.8308 to 0.8875 in 9107 and from 0.7498 to 0.8948 in 9207. The corresponding paired speaker-cluster intervals for channel minus ERM were [0.0296, 0.0842] and [0.1158, 0.1752]. These positive differences were smaller than the development A contrasts. Figure 3 compares the predefined endpoints across partitions. Figure 4 shows the score-level ranking under each held-out condition.

Table 5: Channel minus ERM. Brackets give within-seed 95 percent speaker-cluster intervals. Mean and sample SD describe the three fitted contrasts. Error differences are percentage points.

Endpoint	9107	9207	9307	Mean and SD
A macro stress AUC	+0.1312 [+0.0207, +0.2387]	+0.1893 [+0.0932, +0.2684]	+0.2436 [+0.1884, +0.3071]	+0.1880 ± 0.0562
A clean REAL FA	+0.00 [-2.68, +2.68]	+0.00 [-2.68, +2.68]	+1.79 [-3.57, +9.82]	+0.60 ± 1.03
A clean FAKE miss	+2.68 [+0.00, +6.25]	+0.00 [-3.57, +4.46]	-1.79 [-4.46, +1.79]	+0.30 ± 2.25
A stress REAL FA	-2.68 [-8.63, +4.76]	-28.87 [-37.20, -19.05]	+5.06 [-1.79, +10.42]	-8.83 ± 17.78
A stress FAKE miss	-19.94 [-25.31, -14.58]	-2.68 [-10.42, +3.57]	-43.75 [-47.92, -39.58]	-22.12 ± 20.62
B clean REAL FA	+8.33 [+1.39, +17.36]	-4.17 [-6.94, -0.69]	+6.25 [-1.39, +15.28]	+3.47 ± 6.70
A05 stress FAKE miss	+3.70 [-3.70, +12.70]	+16.67 [+3.70, +29.82]	-11.11 [-16.67, -5.26]	+3.09 ± 13.90
A06 stress FAKE miss	+7.41 [-6.67, +22.92]	+27.78 [+11.76, +43.75]	-31.48 [-43.86, -17.54]	+1.23 ± 30.11

Table 6: Development A by condition. Each cell gives ROC-AUC / REAL false alarms / FAKE misses. Each error count has 112 sources of its class.

Condition	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Clean	0.9872 / 3 / 6	0.9814 / 3 / 9	0.9851 / 4 / 9	0.9809 / 4 / 9
Half gain	0.9812 / 0 / 15	0.9818 / 0 / 14	0.9758 / 0 / 16	0.9820 / 0 / 15
Quarter gain	0.9237 / 7 / 20	0.9576 / 0 / 17	0.9278 / 9 / 20	0.9590 / 0 / 19
Noise5	0.6567 / 1 / 99	0.7793 / 8 / 64	0.5449 / 22 / 77	0.7579 / 13 / 56
Opus64	0.9884 / 6 / 4	0.9819 / 4 / 6	0.9861 / 9 / 5	0.9839 / 5 / 8
Noise to Opus	0.6763 / 48 / 38	0.8108 / 28 / 37	0.6408 / 104 / 13	0.7981 / 44 / 26
Opus to Noise	0.6814 / 5 / 85	0.8178 / 9 / 54	0.5981 / 41 / 53	0.7958 / 13 / 52

Stressed FAKE misses decreased from 1,370 to 587 in 9107 and from 815 to 453 in 9207, each out of 2,496 correlated views from 832 FAKE sources. The rate differences were −31.37 percentage points [−33.41, −29.36] and −14.50 points [−17.38, −11.67]. Stress REAL false alarms changed from 80 to 81 and from 322 to 136, out of 768 views from 256 sources. The respective intervals were [−3.23, +3.76] and [−29.07, −19.03] percentage points. All registered endpoint and attack intervals used 10,000 valid draws with zero invalid draws. These intervals describe speaker resampling within a fitted seed pair.

6.2 Class-Asymmetric Errors and Operating-Point Trade-offs

Table 6 separates the components of the macro stress endpoint. The table and ROC figure detail the first two seeds. The channel arm also improves AUC under each registered stress condition in 9307. Figure 6 shows the full ROC curves underlying those AUCs. Fixed-threshold changes remain heterogeneous. Under Noise5, FAKE misses fall from 99 to 64 in 9107 and from 77 to 56 in 9207. The corresponding REAL false-alarm counts rise from 1 to 8 in 9107 and fall from 22 to 13 in 9207.

The larger 9207 REAL benefit under the two compositions is visible in the per-condition counts. Noise-to-Opus false alarms fall from 104 to 44, while its FAKE misses increase from 13

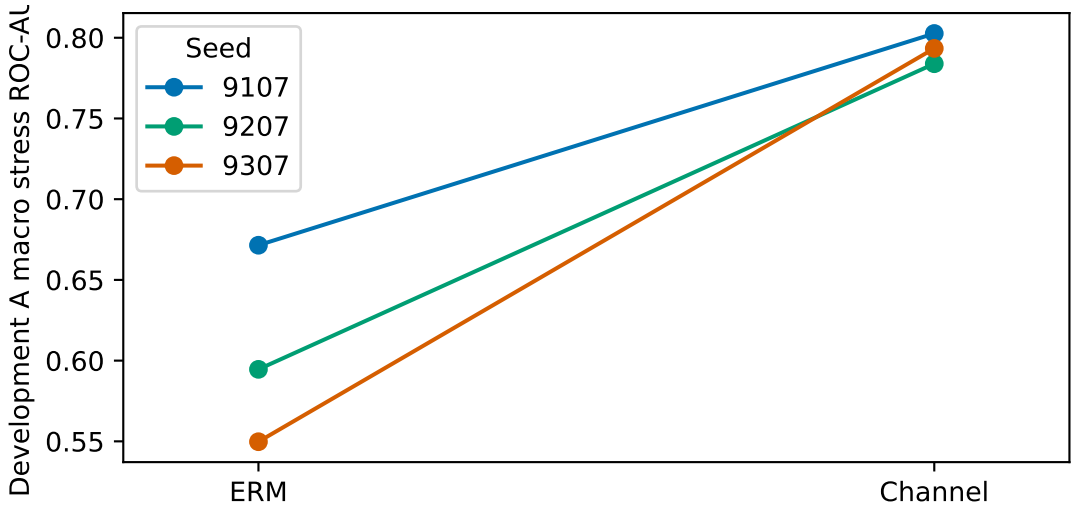


Figure 2: Development macro stress ROC-AUC for the three initialization blocks. Lines connect the two arms within a seed.

Table 7: Selected-model results on the locked held-out subset. Stress error counts pool three views per source. Macro stress ROC-AUC averages the three condition-specific AUCs.

Endpoint	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Macro stress ROC-AUC	0.8308	0.8875	0.7498	0.8948
Stress FAKE miss	1370/2496	587/2496	815/2496	453/2496
Stress REAL false alarm	80/768	81/768	322/768	136/768
Clean REAL false alarm	25/256	19/256	22/256	16/256
Clean FAKE miss	125/832	129/832	127/832	147/832

to 26. In the reverse order, REAL false alarms fall from 41 to 13, while FAKE misses change from 53 to 52. Each count has denominator 112. The baseline's high REAL false-alarm count under Noise-to-Opus provides room for a large reduction at this operating point. It would be misleading to describe that reduction alone as an across-class benefit for the condition.

The gain controls and Opus64 for the first two development seeds provide context for the stress results. Under quarter gain, the channel arm has fewer false alarms and higher AUC in both blocks. Under standalone Opus64, the channel arm has fewer false alarms but more FAKE misses in both blocks. The 64 kb/s control uses a different rate from the 24 kb/s compositions. It is reported as a distinct condition, and any comparison with the chains also changes the codec setting and presence of noise. The matched 24 kb/s diagnostic is the appropriate standalone component comparison for 9107 and 9207.

In 9307, stress FAKE misses fall from 266 to 119 of 336 while REAL false alarms rise from 44 to 61 of 336. This contrasts with fewer stressed REAL false alarms in 9107 and 9207. The 9307 stress miss contrast is −43.75 percentage points (95

In 9207, reductions in other attacks offset 24 additional A05 and A06 stressed misses. Those two attacks contribute six additional misses in 9107. These differences count transformed views. Table 10 reports how many distinct A05 and A06 recordings gain or lose a correct decision in at least one stress condition. Figure 7 extends the paired accounting to both classes across A.

Clean REAL false alarms decreased from 25 to 19 and from 22 to 16 out of 256. Clean FAKE

Table 8: Condition-specific held-out performance. Each cell gives ROC-AUC / REAL false alarms / FAKE misses. There are 256 REAL and 832 FAKE recordings per condition.

Condition	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Clean	0.9534 / 25 / 125	0.9436 / 19 / 129	0.9507 / 22 / 127	0.9410 / 16 / 147
Noise5	0.8053 / 0 / 701	0.8655 / 12 / 259	0.6815 / 28 / 492	0.8765 / 25 / 193
Noise to Opus	0.8447 / 76 / 143	0.8991 / 53 / 143	0.8196 / 226 / 38	0.9044 / 89 / 93
Opus to noise	0.8424 / 4 / 526	0.8979 / 16 / 185	0.7483 / 68 / 285	0.9034 / 22 / 167

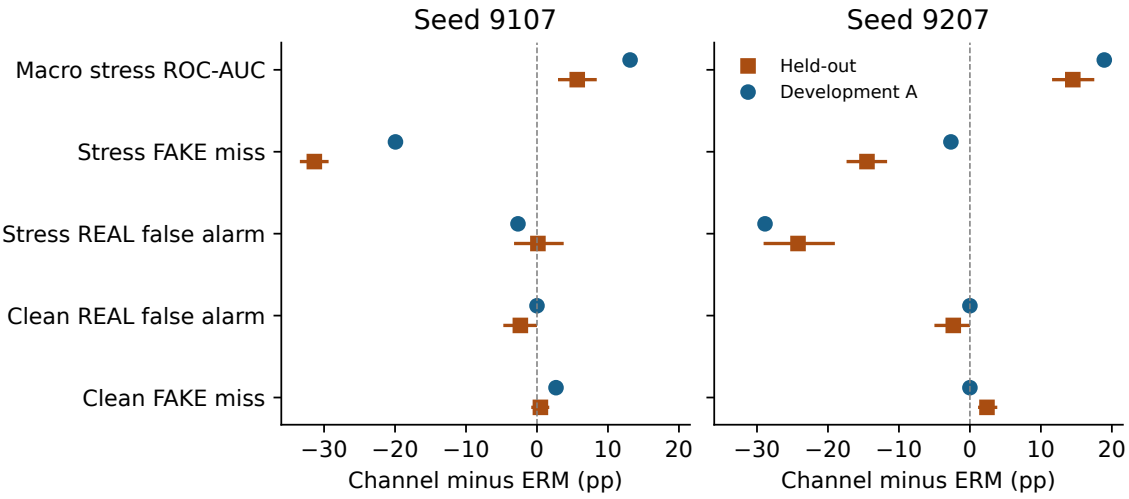


Figure 3: Channel-minus-ERM contrasts on development A and the locked held-out subset, separately by seed. Held-out segments show 95% paired speaker-cluster percentile intervals. AUC differences are multiplied by 100. Positive differences indicate better AUC or higher error rates. Cohorts and attack populations differ across partitions.

misses increased from 125 to 129 and from 127 to 147 out of 832, while clean ROC-AUC declined from 0.9534 to 0.9436 and from 0.9507 to 0.9410. In 9207 the clean FAKE miss increase was 2.40 percentage points [1.15, 3.86]. Both held-out pairs had fewer clean REAL false alarms, whereas the development B contrasts differed in sign. The cohorts have different speaker and source support.

The stress totals conceal an operating-point trade-off. Under Noise5, FAKE misses fell from 701 to 259 and from 492 to 193 out of 832. Under Opus followed by Noise10, they fell from 526 to 185 and from 285 to 167. Under the reverse chain, Noise10 followed by Opus, FAKE misses stayed at 143 in 9107 and rose from 38 to 93 in 9207, even as 9207 REAL false alarms fell from 226 to 89 out of 256 and AUC rose from 0.8196 to 0.9044.

A post-hoc paired decomposition shows what changed at 0.5. Under Noise10 followed by Opus in 9207, the two models shared 85 REAL false alarms, channel corrected 141 and introduced four. For FAKE, the models shared 20 misses, channel corrected 18 and introduced 73. In 9107 the tied total of 143 FAKE misses under that chain comprised 93 shared misses, 50 corrected and 50 introduced. These counts compare decisions on identical sources. They describe why equal totals can conceal a change in the affected recordings.

The retrospective grid in Fig. 8 leaves the registered threshold at 0.5. On held-out 9207 under Noise10 followed by Opus24, channel augmentation gives fewer false alarms and more misses at every sampled threshold from 0.20 through 0.80. The miss direction reverses at 0.10 and 0.90.

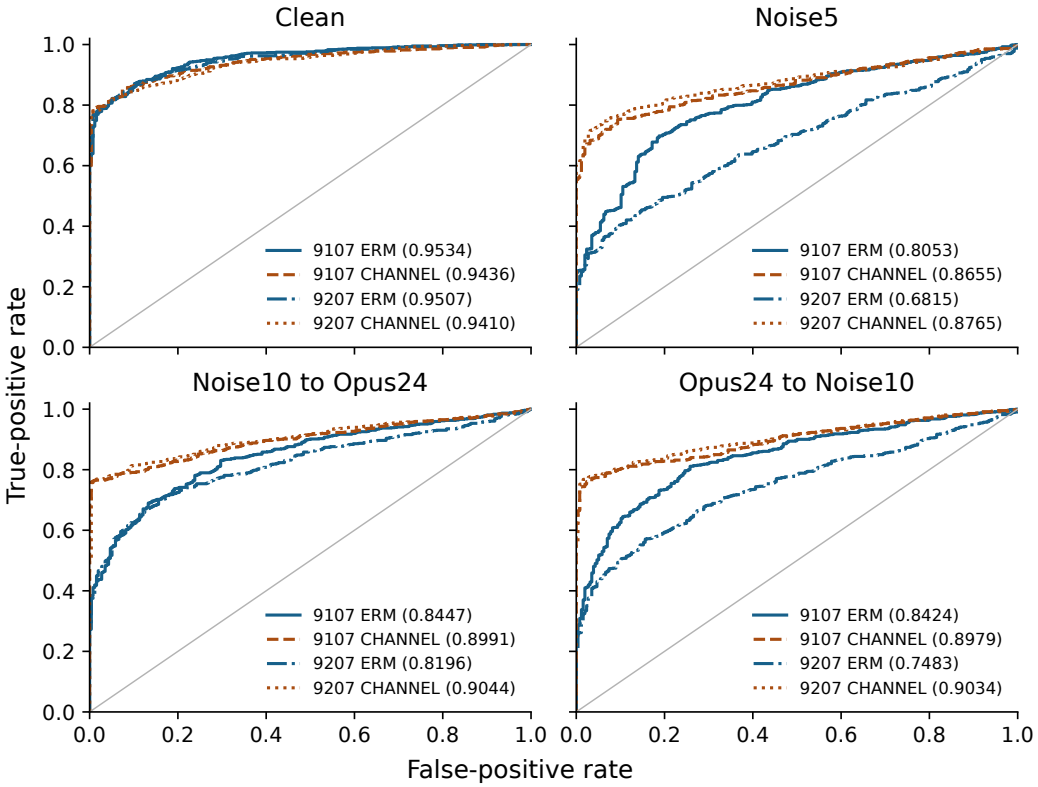


Figure 4: Empirical held-out ROC curves for four selected models. Each panel contains 256 REAL and 832 FAKE recordings, evaluated under the named condition. Legends give seed, method, and AUC. The same sources recur across conditions.

For Noise5 and the opposite order, FAKE misses decrease at all 11 sampled thresholds in both seeds, with condition- and seed-dependent REAL errors. Clean FAKE misses in 9207 increase at all sampled thresholds. These comparisons describe the measured score grid and provide no threshold-selection rule.

Score distributions also change locally around the registered boundary. Under Noise10 followed by Opus24 in 9207, 7 versus 30 of 256 REAL scores and 5 versus 18 of 832 FAKE scores lie within 0.025 of 0.5 for ERM and channel, respectively. The supplement gives the confusion matrices for every arm and condition, empirical curve coordinates and the 0.025 and 0.05 score windows. Development B supplies only REAL scores. Its FNR, ROC-AUC and EER are undefined.

6.3 Attack-Specific Robustness and Failure Modes

Table 9 and Fig. 9 show a sign reversal for both development attacks. Channel increases A05 and A06 stress misses in the first two pairs and decreases them in 9307. The latter changes are $-6/54$ and $-17/54$, respectively. The pointwise 9307 speaker intervals are $[-16.67, -5.26]$ and $[-43.86, -17.54]$ percentage points, respectively. The 9107 intervals include zero. These within-seed intervals do not establish a stable attack effect across initializations.

Table 10 details the first two seeds. In 9307, all six A05 recoveries occur in the noise-before-Opus condition. For A06, 18 recoveries and one new miss occur across the three stress views. Net recoveries span five A05 speakers and six A06 speakers. The A06 ERM burden is 16 of 54 in

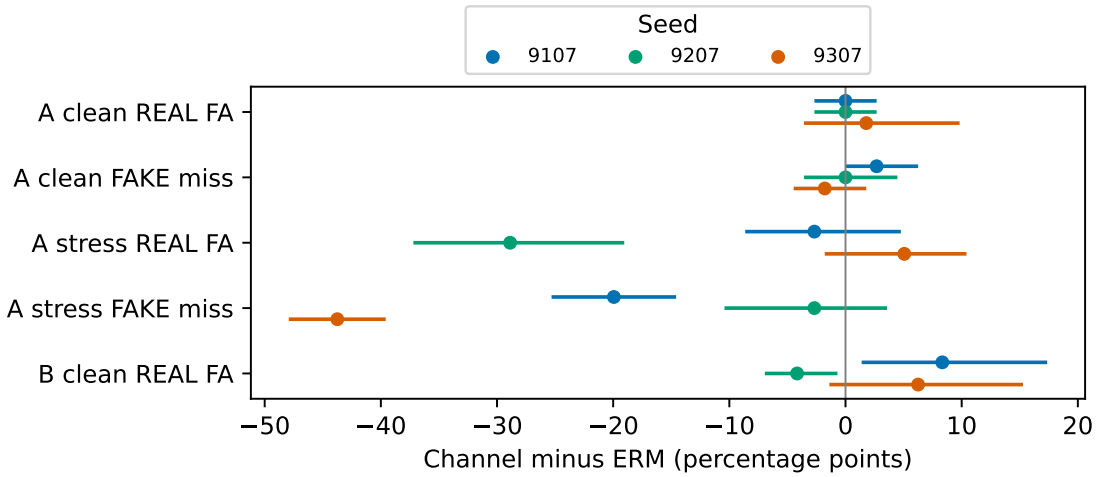


Figure 5: Channel-minus-ERM fixed-threshold error rates by development stratum, class, and seed. Segments show within-seed paired speaker-cluster percentile intervals. REAL errors are false alarms and FAKE errors are misses. Clean A has 112 recordings per class, stress A has 336 views from 112 recordings per class, and clean B has 144 REAL recordings.

Table 9: Development A05/A06 stress FAKE misses over the three stress views. Each row represents 18 distinct sources and 54 transformed views.

Seed and attack	ERM	Channel	Δ misses
9107 A05	48/54	50/54	+2
9107 A06	41/54	45/54	+4
9207 A05	38/54	47/54	+9
9207 A06	16/54	31/54	+15
9307 A05	49/54	43/54	-6
9307 A06	45/54	28/54	-17

9207 and 45 of 54 in 9307. Differences in baseline burden and FAKE score crossings describe where the reversal occurs. They leave the underlying model or generator mechanism unresolved.

Figure 10 shows that two B speakers contribute 10 of the 12 net additional false alarms in 9107. For 9207, reductions occur across several speakers. B contains only REAL recordings, and the counts are interpreted as false alarms with no B ROC-AUC or EER.

Figure 11 shows the condition-specific changes for A07–A19. Stress misses declined on A07–A16 in both blocks. They increased on A17, A18 and A19 in both blocks, by 10, 11 and four views in 9107 and by 43, 43 and 56 views in 9207. Each attack contributes 64 sources and 192 stressed views. The 9107 intervals for these three increases include zero, while the 9207 intervals are [15.90, 29.03], [14.12, 30.81] and [20.63, 37.70] percentage points. These descriptive attack contrasts expose adverse subgroups alongside the pooled improvement. The development A05/A06 observations and held-out A17–A19 observations refer to different attack sets.

In the post-hoc breakdown, A07–A16 contribute net reductions of 808 and 504 stressed misses, partially offset by increases of 25 and 142 on A17–A19. In 9107 the adverse net increases are driven by Noise10 followed by Opus, which adds 13 misses to each group. In 9207 all three stress conditions add misses on these attacks. For A19, 43 of 64 distinct sources acquire at least one new stressed miss and 18 acquire new misses under multiple conditions in 9207. These groupings were identified after examining the held-out results. The intervals are pointwise and

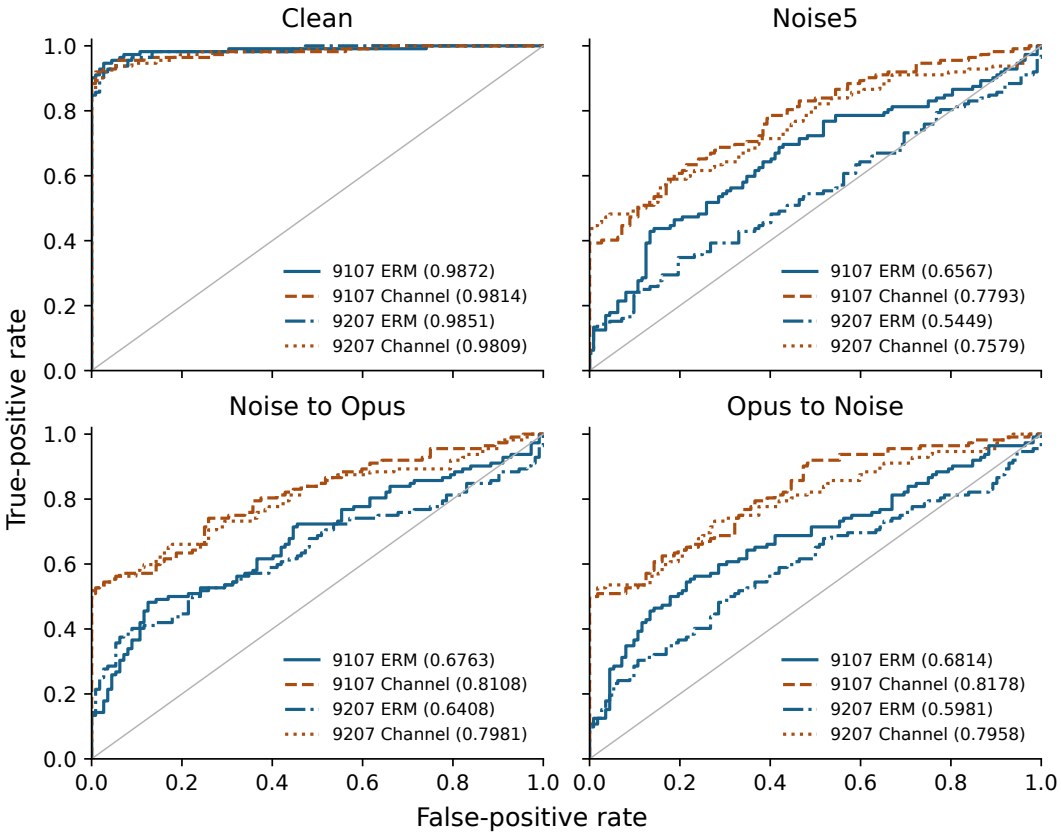


Figure 6: Empirical ROC curves from the four selected models in the first two seeds on development A. Each panel ranks the same 112 REAL and 112 FAKE recordings under one condition. Legends give the seed, method, and recomputed AUC. The diagonal denotes chance ranking.

are interpreted descriptively.

6.4 Transformation-Order Effects

In the first two development blocks, among 112 A REAL sources, the Noise-to-Opus-only false-alarm count changes from 43 to 19 in 9107 and from 63 to 31 in 9207. The opposite order has no exclusive REAL false alarms in A. Among 112 A FAKE sources, Opus-to-Noise alone causes 48 versus 17 misses in 9107 and 40 versus 26 in 9207. The Noise-to-Opus-only FAKE counts are 1 versus 0 and 0 versus 0. The 9307 pair shows the same classwise order asymmetry. Noise-to-Opus-only REAL errors change from 37 to 44, and Opus-to-Noise-only FAKE misses change from 41 to 33, each out of 112. The median signed FAKE score gap falls from 0.2375 to 0.0538. Figure 12 shows the paired score contrasts before the threshold is applied.

A retrospective 9107 diagnostic adds 1,472 predictions for standalone Noise10 and Opus24. Matched waveforms reconstruct both registered chains and selected anchor scores replay without observed drift. A leave-one-speaker-out nuisance regression yields directional class-discrimination AUCs of 0.538 for ERM and 0.432 for channel on the signed order score. This measures a residual score contrast on the development cohort. The same standalone components were evaluated for seed 9207 on all 368 development sources, adding 1,472 predictions. All 18 clean and ordered score anchors matched their sealed values exactly. On development A, standalone Noise10 FAKE

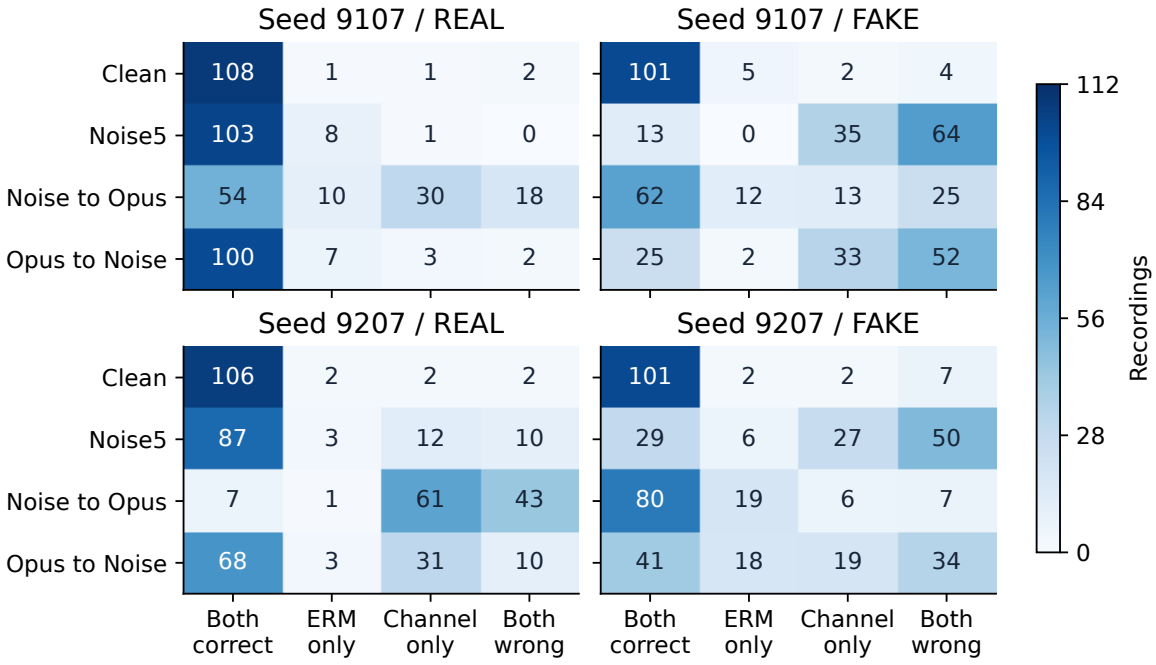


Figure 7: Within-source correctness for the first two seeds at threshold 0.5 on development A. Each row contains 112 recordings of the indicated class. The same recordings contribute to each condition row. The two middle columns identify sources correctly classified by only one method.

Table 10: A05 and A06 source-level diagnostics. Each condition has 18 sources. Error cells are ERM / channel. Harm and recovery can overlap across views.

Seed / attack	Clean	Noise5	Noise to codec	Codec to noise	All stress	Harmed / rec.
9107 / A05	0 / 0	17 / 17	14 / 16	17 / 17	14 / 16	3 / 1
9107 / A06	3 / 6	18 / 16	7 / 12	16 / 17	6 / 12	7 / 2
9207 / A05	0 / 0	17 / 17	7 / 13	14 / 17	7 / 13	8 / 0
9207 / A06	5 / 7	9 / 12	0 / 5	7 / 14	0 / 5	10 / 0

misses decreased from 82 to 54 in 9107 and changed from 47 to 48 in 9207, out of 112 per arm. Under standalone Opus24, 9207 REAL false alarms decreased from 19 to five and FAKE misses increased from two to ten. The 9107 nuisance regression remains a single-seed analysis. These component results constrain interpretation of the compositions and do not identify a forensic mechanism.

Source-linked order asymmetry persisted. In 9107, REAL sources wrong only under noise-to-Opus numbered 72 for ERM and 37 for channel, with zero exclusive errors in the reverse direction. In 9207 the corresponding counts were 158 and 68, while exclusive reverse-order errors numbered zero and one. Among FAKE sources, reverse-order-only misses numbered 384 and 43 in 9107 and 247 and 74 in 9207. The opposite exclusive FAKE miss counts were one and one in 9107, and zero and zero in 9207. Each model uses the same 256 REAL and 832 FAKE sources in the order comparison.

The score contrasts in Fig. 13 also remained directional, with lower FAKE-class medians after channel training. Median FAKE contrasts changed from 0.3454 to 0.0063 in 9107 and from 0.2419 to 0.0076 in 9207. Smaller score contrasts alone do not imply a uniformly better decision rule, as shown by the attack-specific misses. Speaker-linked outputs retain the distribution of

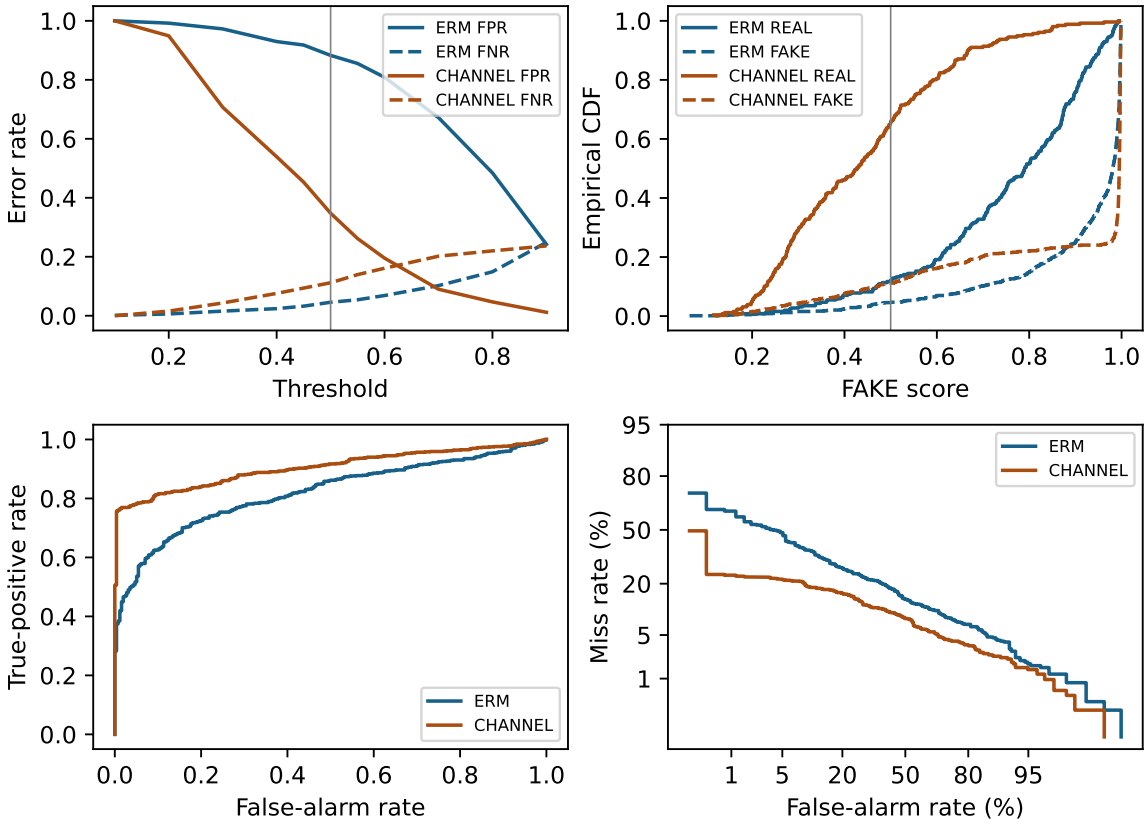


Figure 8: Held-out seed 9207 under Noise10 followed by Opus24. Top left shows exploratory FPR and FNR at 11 thresholds. Top right shows class-specific empirical CDFs. Bottom panels show empirical ROC and DET. DET axes use a normal quantile transform and clip endpoint rates at half an event per class. The vertical lines identify the registered threshold of 0.5. Each arm has 256 REAL and 832 FAKE sources.

order errors. For example, ERM 9107's reverse-minus-forward FAKE error differences were 16 among 21 sources for LA_0045 and 13 among 21 for LA_0046. These speaker counts describe concentration within the selected corpus and support no demographic inference.

6.5 Robustness, Stability and Generalization

The direction of the development stress-ranking gain and pooled FAKE miss reduction repeats in all three pairs. The size of each gain and the direction of false-alarm effects depend on the fitted pair. A shared development cohort supports a comparison of initializations under this protocol, with speaker uncertainty measured within each pair.

The locked held-out result tests the original four models on new speakers within ASVspoof 2019 LA. Seed 9307 contributes development evidence only. Cross-corpus robustness and sensitivity to training size remain open.

6.6 Practical Implications and Limitations

In 9307 the recorded optimizer durations were 39.9 minutes for ERM and 78.9 minutes for channel. Validation took approximately 37.7 and 37.8 minutes, respectively. Peak resident memory was approximately 714 and 896 MiB. These are CPU workload observations for the medium cohort.

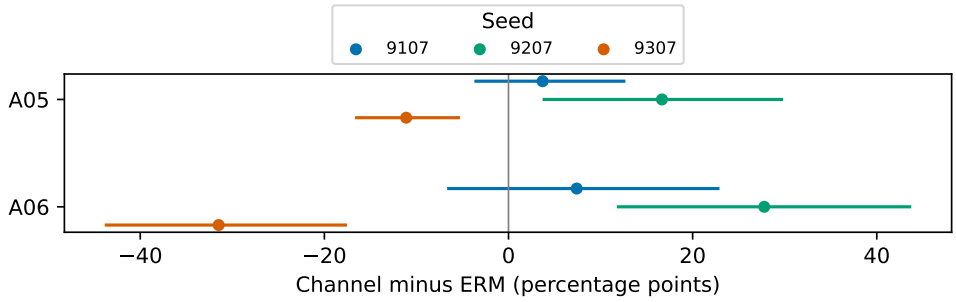


Figure 9: Development A05/A06 stressed FAKE miss changes by seed. Positive values indicate more misses. Intervals resample speakers within a seed. Each attack contributes 54 views from 18 recordings.

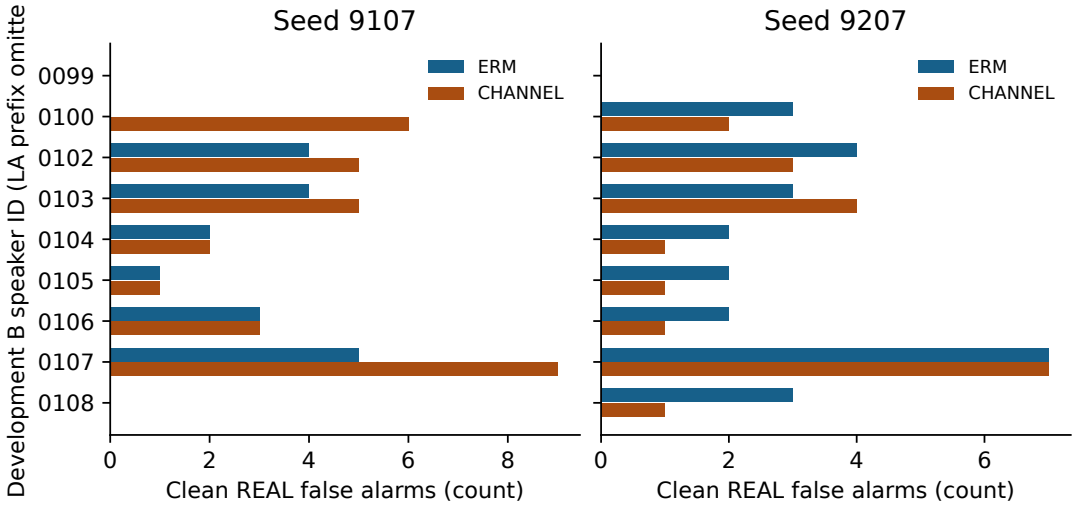


Figure 10: Clean REAL false alarms by speaker on development B for the first two seeds. Each of nine speakers contributes 16 recordings, for 144 recordings per arm and seed.

They do not quantify large-cohort costs.

A bounded inference benchmark used the clean inputs of the first 12 registered development sources, repeated three times after anchor replay, at batch size one on a single CPU thread. Input duration was capped at 3 s. Median preprocessing time was 10.9 ms (interquartile range 10.3–11.2 ms), forward time 91.3 ms (90.5–93.5 ms), and end-to-end time 102.1 ms (101.1–104.2 ms). The timed preprocessing includes decoding, resampling and the P0 input transform on clean input. The separate headroom and channel-transform stages are outside the timed path. The two 9207 exports each contain 1,308,401 parameters, of which 369,281 are updated during warm-up and 646,481 during partial fine-tuning. The exported stage has 661,920 nontrainable parameters. Each export occupies 5,916,699 bytes. Load times in one process were 3.35 and 2.81 s, with the second load following the first. Peak process working set through the timing snapshot was approximately 475 MiB. The development convenience sample supports an architectural and local runtime description. Mobile, streaming and concurrent performance remain unmeasured.

The error curves illustrate why a new operating point requires independent validation with costs and traffic prevalence appropriate to the deployment. The ordered chains further motivate validating the location at which noise is introduced relative to codec processing.

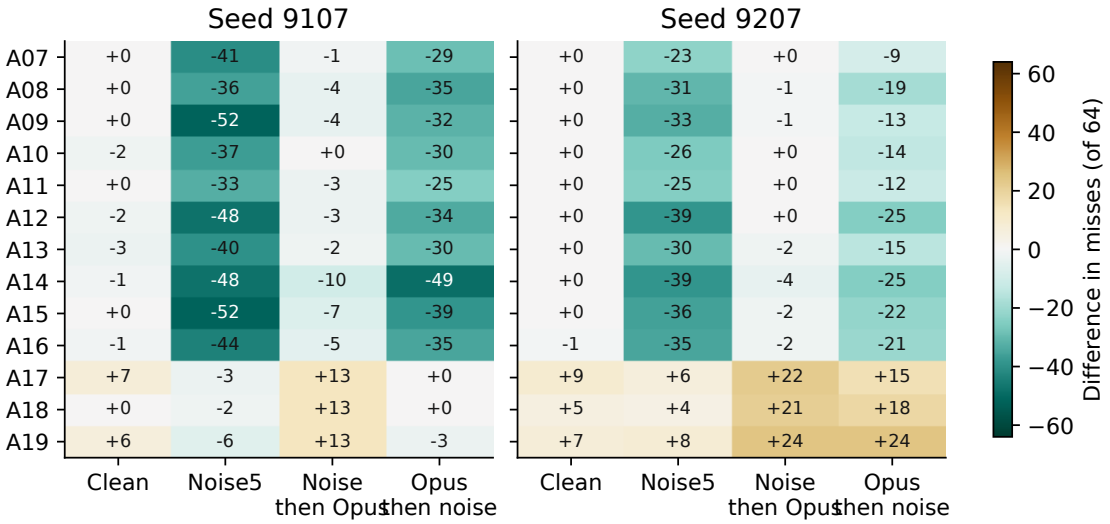


Figure 11: Held-out channel-minus-ERM FAKE miss counts by attack, condition and seed. Each cell compares the same 64 sources. Positive counts indicate deterioration and negative counts indicate fewer misses. The full numerical supplement retains both arm counts and stressed attack intervals.

7 Discussion

The stress-ranking gain and pooled stress FAKE miss reduction repeat across the three development seed blocks. A05 and A06 show that the aggregate direction can coexist with attack effects that reverse across fitted pairs. The third block therefore strengthens the evidence for the aggregate contrast under the registered protocol and narrows the claim about attack-level behavior.

Clean B false alarms increase in 9107 and 9307 and decrease in 9207. Stressed REAL false alarms also reverse direction in 9307. The results describe decisions under the fixed threshold. They motivate reporting the two class risks separately when ranking improves.

For many REAL sources, Noise-to-Opus is the only erroneous order. For FAKE sources, the reverse chain is often the only erroneous order. An upward shift in FAKE scores under one order can produce both patterns. The retrospective matched-component diagnostic for 9107 finds that nuisance adjustment does not establish an independent forensic target. Nonlinear codec response, preprocessing, and detector sensitivity may all contribute. Reproducing the direction of decision asymmetry in 9207 and 9307 does not independently reproduce the nuisance analysis.

In practical evaluation, reporting both threshold-specific risks and ranking separates several failure modes that may have different costs. Speaker concentration also affects how a review workload might change, although no field workload or utility was measured. Conventional Opus order reversal, class-wise counts and a matched training intervention provide a focused complement to broader robustness benchmarks and codec-adaptation methods.

7.1 Aggregate improvement and its distribution

The matched comparison makes training-view policy the controlled factor, but does not isolate a single cause inside the fitted network. The channel model can learn features that distinguish

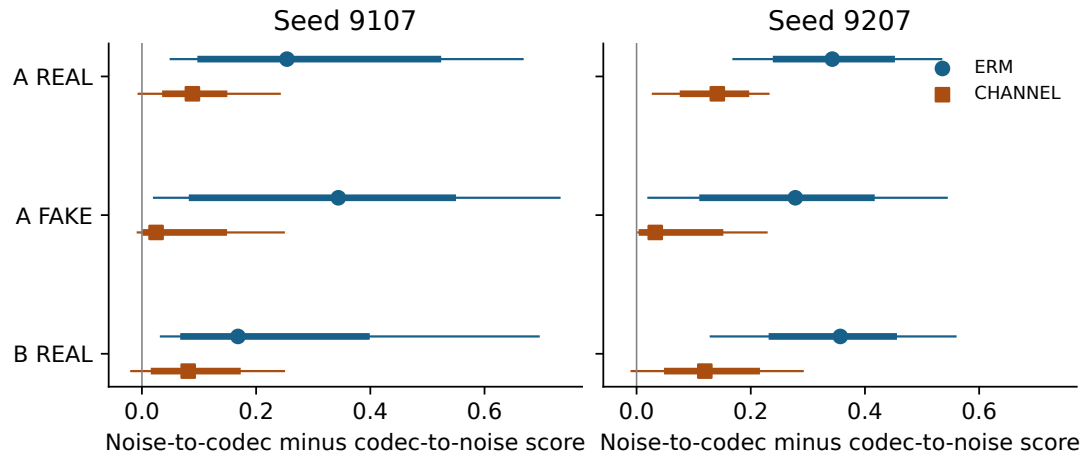


Figure 12: Directional score contrasts for the first two development seeds on the same recording in opposite processing orders. Points are medians, thick segments span quartiles, and thin segments span the 10th to 90th percentiles. Each A class has 112 sources and B has 144 REAL sources. Segments describe score distributions.

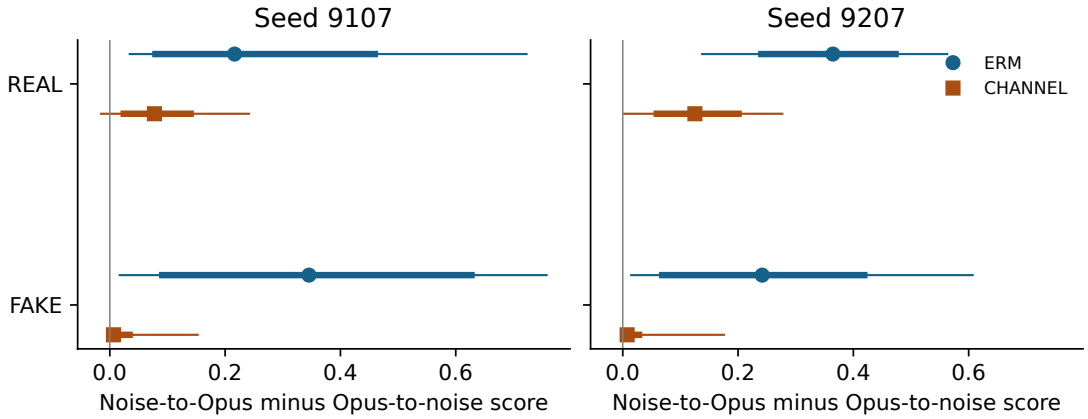


Figure 13: Held-out same-source FAKE score contrasts between noise-to-Opus and Opus-to-noise. Points are medians, thick segments span quartiles, and thin segments span the 10th to 90th percentiles. REAL and FAKE groups contain 256 and 832 sources. These are distribution summaries of paired scores.

more stress-processed pairs, move score distributions relative to the fixed decision boundary, or do both. Clean BCE selection may choose a different epoch in each arm because the fitted score distributions are different. The observed contrast therefore includes the interaction of augmentation with the same selection rule. It cannot determine which intermediate representation would be responsible, or how an alternate selection rule would change outcomes.

A05 and A06 each have 18 development sources, and an individual source can contribute up to three stress errors. The 9307 source-linked changes involve several speakers and different conditions across attacks. A higher baseline miss burden leaves room for more recoveries, but the available scores and speaker counts do not isolate the source of a seed-dependent reversal.

REAL false alarms can shift if a score displacement moves bona fide examples near 0.5 across the threshold. In B the three blocks have mixed directions, and part of the 9107 increase concentrates in a small number of speakers. Bona fide cross-testing work makes the broader case

for assessing bona fide diversity [3]. Our speaker analysis addresses a narrower issue inside one fixed cohort. A different bona fide corpus, language, noise mixture or deployment prior could change both the baseline rates and the direction of the intervention effect.

7.2 Signal-chain and engineering implications

Published communication benchmarks such as ADD-C vary codec and packet-loss conditions [2]. Their breadth is useful for establishing stress coverage. The same-source order contrast here trades breadth for experimental control. It compares two explicitly defined pipelines at fixed settings and records exactly which order makes a particular decision fail. The association can motivate a follow-up test with actual transmission and varied gain controls. It cannot stand in for that test because the present experiment omits network loss and device processing.

Where a review decision is threshold based, a deployment report should specify the order and settings of the signal-processing chain and give error counts by class. A rank statistic can summarize detector ordering, and an error transition table can show whether a system has shifted errors to a different group. The latter is valuable even if total counts tie. For example, two methods can flag the same number of bona fide sources but require review of different recordings. No application-specific cost or calibration was measured, so a cost-optimal operating rule remains a separate problem.

The present controlled intervention uses ordinary BCE and one view per source. Its scientific contribution is the error accounting and the evidence that the observed rank benefit coexists with subgroup deterioration. Extending the claim would require a locked independent partition, then genuinely new corpora or models to measure broader transport. Confirmation of a forensic mechanism would additionally require a cue definition and a measurement that can vary independently of the detector output.

7.3 From a validation metric to a validation workflow

For an engineer deciding whether to adopt augmentation, the condition table distinguishes several cases. An AUC increase with reductions in both error types shows improvement at both levels measured. An AUC increase with fewer false alarms and more misses indicates that a rank gain and an operating-point trade-off coexist. A near-tie in counts can still conceal a change in the individual sources that require attention. The present results exhibit these behaviors under different conditions. A validation workflow benefits from retaining source identity through the paired analysis.

Nominal condition names can also obscure the processing intervention. A label that names noise and Opus omits whether the noise is present during coding. The operator descriptions here fix that detail and define how the noise realization is shared. This isolates sensitivity to the specified order without assuming either chain represents all communication paths. It also motivates reporting codec rates alongside order. Opus64 has different settings and support from the 24 kb/s compositions, and their contrast cannot isolate bitrate. The present evidence pertains to a classifier after specified transformations. A claim about forensic cue retention would require a separate measurement that varies independently of classifier output.

7.4 Evidence needed to extend the claim

The conference reviewers raised related questions about broader robustness and repeatability. The journal makes progress by separating training initialization from test-source resampling, documenting how the operating rule interacts with class-specific effects, and retaining source links through the analysis. These changes address inference and diagnosis within the registered scope. They leave broader shifts unresolved. The locked within-release evaluation extends the positive stress-ranking contrast to new speakers and a different attack set. External-corpus, physical-channel, and multi-architecture transfer remain open.

The held-out ranking differences are smaller than the development differences, while the pooled FAKE miss reductions are larger. The different populations and denominators preclude a controlled estimate of the effect of changing partition. The A05 and A06 development errors cannot be projected directly onto A07 through A19. The official corpus documentation relates some attack algorithms across partitions, and the five-column metadata does not identify full generator lineage. The scientific conclusion should therefore be phrased in terms of the designated partitions and conditions. Claims of genuinely unseen generator robustness require an additional design with independently verified generator provenance.

8 Limitations and Threats to Validity

The balanced 640-source cohort is a compute-bounded training design. Full-foundation training was outside the design and would change the class mixture from 50% FAKE to 89.83% FAKE. A and B have limited speaker support, and development attack groups have 18 or 19 sources. Related conditions within a source are dependent. The bootstrap resamples speakers with the models fixed. It does not characterize between-initialization variability.

The three development initialization blocks share the same development data and checkpoint rule. The same partition supports checkpoint selection and the reported development analysis. The results are sensitive to the fixed decision threshold and selected checkpoint. In particular, guard failure applies to the selected checkpoint comparison. It does not imply that every epoch or threshold would fail.

Only one architecture was retrained in the matched design. Signal preprocessing, noise scaling, and codec response can change the score contrast independently of forensic cue retention. The current experiments provide no verified independent forensic target. The completed official evaluation is within the same corpus release, and some evaluation attacks reuse algorithms or frameworks. Cross-corpus, cross-architecture, and deployment robustness remain open.

8.1 Internal and external validity

The signal intervention controls the configured processing and uses matched inputs, but the clean cohort determines which speakers and synthesis examples can contribute. Within release, reuse of attack families prevents a simple unseen-generator interpretation. Further, the detector selects checkpoints on clean development BCE, which can couple the reported clean performance to the selection data. The registered selection prevents retrospective adjustment to the stress effects

while leaving this dependence intact. Effects on other operating points and threshold policies would need a separate locked design.

The source is the unit for linking ordered views, and the speaker is the resampling unit. A model sees correlated transformed versions of the same speech signal at evaluation. A larger view count does not replace additional speakers. In particular, attack-level intervals depend on a limited number of source and speaker clusters. A different bootstrap specification, broader cohort, or leave-one-speaker-out analysis could change interval width. The current inference is limited to the reported conditional comparisons.

8.2 Engineering scope

Execution integrity, deterministic replay checks and sealed artifacts support reproducibility of the fitted comparisons. They do not establish that a production implementation has the same numerical or latency properties. The measurements use a CPU with the registered environment and include training and validation work. They provide no evidence about mobile latency, energy, streaming, or adversarial adaptation. The completed evaluation uses a bounded, attack-balanced within-release subset. Representative codecs, networks, devices, languages, and authentic speech conditions require separate measurements.

8.3 Construct validity and paired uncertainty

A FAKE score is a classifier output. It can change when the transformation affects spectral structure, temporal texture, signal scale or the preprocessing input range. Without an independently defined cue and intervention, the change cannot distinguish altered evidence from altered sensitivity to the same evidence. The matched components provide a check against an overly simple additive interpretation, while the one-block retrospective scope limits replication.

The within-seed paired estimates reflect a common test population and depend on the registered speaker clusters. The selected clean checkpoint is another source of uncertainty because different realizations of the training corpus or selection data could select another epoch. The observed differences across three blocks are useful robustness diagnostics but cannot estimate a reliable distribution over seeds, architectures or corpora. The manuscript therefore reports each development contrast separately and treats their mean and sample SD as descriptive. The locked held-out evidence covers the original two pairs.

The held-out sample contains 64 sources per FAKE attack and 256 REAL sources from 67 speakers. This quota design changes attack weighting relative to the complete official partition. The attack intervals are pointwise and do not control multiplicity. Highlighting A17–A19 and decomposing specific errors were post-hoc choices, so subgroup patterns remain exploratory even where individual intervals exclude zero. These recordings were first evaluated with a locked design. Subsequent model selection on their scores would consume that separation. Generalization across corpora, generators, training scales and architectures remains unresolved.

9 Conclusion

Channel augmentation improved stressed ranking and reduced pooled stressed FAKE misses in all three matched development blocks and in the original two blocks on the locked within-release evaluation. The direction of stress REAL false alarms and A05/A06 misses varied across development seeds. Clean FAKE sensitivity declined on held-out, and the descriptive attack analysis found additional stressed misses on A17–A19. Under noise followed by Opus in 9207, channel reduced REAL false alarms while increasing FAKE misses at the same operating point.

Reversing noise and codec order produced different REAL and FAKE error transitions on the same held-out recordings. The measured behavior supports reporting rank metrics alongside classwise risks and testing the two channel orders when they are relevant to the application. The evidence concerns one architecture, three development seed blocks, two locked held-out blocks and one corpus release. Transfer beyond that scope and the forensic mechanism underlying the response remain unresolved.

Reproducibility

The experiment registry, per-source scores, checkpoint seals, and speaker-bootstrap outputs document the development and held-out comparisons. The accompanying analysis code checks the locked manifest and score identities, reproduces the registered metrics for six development models and four locked held-out models, and derives the post-hoc threshold, attack and order diagnostics from the recorded predictions. The supplement provides per-seed arm metrics, paired speaker intervals, descriptive across-seed statistics and the original V4 threshold and matched-component diagnostics. Section 5 defines the evaluation populations.

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References

[1] H. Tak, M. Kamble, J. Patino, M. Todisco, and N. Evans, "RawBoost: A Raw Data Boosting and Augmentation Method applied to Automatic Speaker Verification Anti-Spoofing," in Proc. *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 6382–6386 (2022), doi:10.1109/ICASSP43922.2022.9746213, `https://www.eurecom.fr/en/publication/6736`.

[2] H. Shi, X. Shi, S. Dogan, et al., "Benchmarking Audio Deepfake Detection Robustness in Real-World Communication Scenarios," in Proc. *European Signal Processing Conference*, pp. 566–570 (2025), doi:10.23919/EUSIPCO63237.2025.11226601, `https://eusipco2025.org/wp-content/uploads/pdfs/0000566.pdf`.

[3] C. Y. Kwok, J. Q. Yip, Z. Qiu, C. H. Chi, and K. Y. Lam, "Bona Fide Cross Testing Reveals Weak Spot in Audio Deepfake Detection Systems," in Proc. *Interspeech*, pp. 2230–2234 (2025), doi:10.21437/Interspeech.2025-172, `https://www.isca-archive.org/interspeech_2025/kwok25_interspeech.html`.

[4] H. Maltby, J. Wall, C. Glackin, et al., "Robustness in Deepfake Speech Detection: A Survey of Failure Mechanisms Including an Experimental Case Study," *Computer Speech & Language*, vol. 102, p. 102037 (2026), doi:10.1016/j.csl.2026.102037, https://repository.uwl.ac.uk/id/eprint/15273/.

[5] H. Tak, J. Patino, M. Todisco, et al., "End-to-end anti-spoofing with RawNet2," in Proc. *IEEE International Conference on Acoustics, Speech and Signal Processing* (2021), doi:10.1109/ICASSP39728.2021.9414234, https://doi.org/10.1109/ICASSP39728.2021.9414234.

[6] J. weon Jung, H.-S. Heo, H. Tak, et al., "AASIST: Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks," in Proc. *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 6367–6371 (2022), doi:10.1109/ICASSP43922.2022.9747766, https://doi.org/10.1109/ICASSP43922.2022.9747766.

[7] A. Howard, M. Sandler, G. Chu, et al., "Searching for MobileNetV3," in Proc. *IEEE/CVF International Conference on Computer Vision* (2019), https://openaccess.thecvf.com/content_ICCV_2019/html/Howard_Searching_for_MobileNetV3_ICCV_2019_paper.html.

[8] L. Pham, D. Tran, P. Lam, et al., "DIN-CTS: Low-Complexity Depthwise-Inception Neural Network with Contrastive Training Strategy for Deepfake Speech Detection," in Proc. *European Signal Processing Conference*, pp. 556–560 (2025), https://eurasip.org/Proceedings/Eusipco/Eusipco2025/pdfs/0000556.pdf.

[9] X. Wang, J. Yamagishi, M. Todisco, et al., "ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech," *Computer Speech & Language*, vol. 64, p. 101114 (2020), doi:10.1016/j.csl.2020.101114, https://doi.org/10.1016/j.csl.2020.101114.

[10] J. Yamagishi, X. Wang, M. Todisco, et al., "ASVspoof 2021: accelerating progress in spoofed and deepfake speech detection," in Proc. *ASVspoof Workshop*, pp. 47–54 (2021), doi:10.21437/ASVSPOOF.2021-8, https://www.isca-archive.org/asvspoof_2021/yamagishi21_asvspoof.html.

[11] X. Wang, H. Delgado, H. Tak, et al., "ASVspoof 5: Design, Collection and Validation of Resources for Spoofing, Deepfake, and Adversarial Attack Detection Using Crowdsourced Speech," *Computer Speech & Language*, vol. 95, p. 101825 (2026), doi:10.1016/j.csl.2025.101825, https://www.eurecom.fr/publication/8080/download/sec-publi-8080.0-s0885230825000506-main.pdf.

[12] N. M. Müller, P. Czempin, F. Dieckmann, A. Froghyar, and K. Böttinger, "Does Audio Deepfake Detection Generalize?" in Proc. *Interspeech*, pp. 2783–2787 (2022), doi:10.21437/Interspeech.2022-108, https://www.isca-archive.org/interspeech_2022/muller22_interspeech.pdf.

[13] N. M. Müller, P. Kawa, W. H. Choong, et al., "MLAAD: The Multi-Language Audio Anti-Spoofing Dataset," in Proc. *International Joint Conference on Neural Networks*, pp. 1–7 (2024), doi:10.1109/IJCNN60899.2024.10650962, https://www.resemble.ai/resources/mlaad-the-multi-language-audio-anti-spoofing-dataset.

[14] J.-M. Valin, K. Vos, and T. B. Terriberry, "Definition of the Opus Audio Codec," (2012), doi:10.17487/RFC6716, https://www.rfc-editor.org/rfc/rfc6716, rFC 6716.

[15] Z. Zhang, W. Hao, A. Sankoh, et al., "I Can Hear You: Selective Robust Training for Deepfake Audio Detection," in Proc. *International Conference on Learning Representations* (2025), https://proceedings.iclr.cc/paper_files/paper/2025/hash/1ab925f5f4ac0670256c3d335868c996-Abstract-Conference.html.

[16] S. Sagawa, P. W. Koh, T. B. Hashimoto, and P. Liang, "Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization," in Proc. *International Conference on Learning Representations* (2020), https://openreview.net/forum?id=ryxGuJrFvS.

[17] T. M. Wani and I. Amerini, "Dynamic Knowledge Condensation with Audio-Selective Transformer for Audio Deepfake Detection," *Discover Computing*, vol. 28, p. 231 (2025), doi:10.1007/s10791-025-09746-4, https://link.springer.com/article/10.1007/s10791-025-09746-4.

[18] I. N. Hidekel, G. Lifshitz, K. Cohen, and D. Raviv, "SONAR: Spectral-Contrastive Audio Residuals for Generalizable Deepfake Detection," in Proc. *International Conference on Machine Learning*, vol. 306, pp. 42998–43012 (2026), https://proceedings.mlr.press/v306/hidekel26a.html.

[19] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," in Proc. *International Conference on Machine Learning*, vol. 70, pp. 1321–1330 (2017), https://proceedings.mlr.press/v70/guo17a.html.

[20] Y. Geifman and R. El-Yaniv, "SelectiveNet: A Deep Neural Network with an Integrated Reject Option," in Proc. *International Conference on Machine Learning*, vol. 97, pp. 2151–2159 (2019), https://proceedings.mlr.press/v97/geifman19a.html.

[21] O. Pascu, A. Stan, D. Oneata, E. Oneata, and H. Cucu, "Towards Generalisable and Calibrated Audio Deepfake Detection with Self-Supervised Representations," in Proc. *Interspeech*, pp. 4828–4832 (2024), doi:10.21437/Interspeech.2024-1302, https://www.isca-archive.org/interspeech_2024/pascu24_interspeech.html.

[22] H. Tak, M. Todisco, X. Wang, et al., "Automatic Speaker Verification Spoofing and Deepfake Detection Using Wav2vec 2.0 and Data Augmentation," in Proc. *The Speaker and Language Recognition Workshop (Odyssey 2022)*, pp. 112–119 (2022), doi:10.21437/Odyssey.2022-16, https://www.isca-archive.org/odyssey_2022/tak22_odyssey.html.

[23] I.-P. Ciobanu, A.-I. Hîji, N. C. Ristea, et al., "XMAD-Bench: Cross-Domain Multilingual Audio Deepfake Benchmark," in Proc. *Findings of the Association for Computational Linguistics: EACL 2026*, pp. 3109–3120 (2026), doi:10.18653/v1/2026.findings-eacl.162, https://aclanthology.org/2026.findings-eacl.162/.

[24] J. A. Hanley and B. J. McNeil, "The Meaning and Use of the Area Under a Receiver Operating Characteristic (ROC) Curve," *Radiology*, vol. 143, pp. 29–36 (1982), doi:10.1148/radiology.143.1.7063747, https://doi.org/10.1148/radiology.143.1.7063747.

[25] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in Proc. *International Conference on Learning Representations* (2015), https://arxiv.org/abs/1412.6980.

[26] C. A. Field and A. H. Welsh, "Bootstrapping Clustered Data," *Journal of the Royal Statistical Society: Series B*, vol. 69, pp. 369–390 (2007), doi:10.1111/j.1467-9868.2007.00593.x, https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9868.2007.00593.x.